In-Class Assignment L2-1: Running Time Analysis (Group of 2 to 3)

(25 minutes) Submit to LEB2

From the algorithm below,

- What task does it solve ?
- What is the basic operation in the algorithm ?
- Are there best-case and worst-case $C(n)$? When do they happen?

  If so, determine $C(n)$ in each scenario.

Algorithm

```
1:  Input: Array A[1 ... n] of integers
2:  Output:
3:
4:  for i = 2 to n do
5:      v ← A[i]
6:      j ← i − 1
7:      while j ≥ 1 and A[j] > v do
8:          A[j + 1] ← A[j]
9:          j ← j − 1
10:     A[j + 1] ← v
```